Cache-Line- and SIMD-Aware Scan

Let the two-dimensional plane be

$$A \in \mathbb{R}^{M \times N}, \qquad A_{i,j}, \quad i = 1, \ldots, M, \quad j = 1, \ldots, N,$$

where i indexes rows and j indexes columns.

Hardware model

Let

$$B = \text{cache-line size in bytes}, \qquad s = \text{bytes per element},$$

so that one cache line contains

$$E = \frac{B}{s}$$

elements. Let w be the SIMD width, i.e. the number of elements processed simultaneously by one vector instruction. Assume

$$w \mid E, \qquad V = \frac{E}{w},$$

so that each cache line contains V SIMD vectors.

Because the memory system transfers entire cache lines, columns are processed in cache-line blocks. Rows are padded to a cache-line boundary:

$$N_p = \left\lceil \frac{N}{E} \right\rceil E, \qquad L = \left\lceil \frac{N}{E} \right\rceil,$$

where L is the number of cache lines touched by one row. The padding columns are never counted in the average.

Row groups

Partition the rows into consecutive groups

$$\{1, \ldots, M\} = \bigcup_{k=1}^{K} G_k, \qquad G_k = \{r_{k-1} + 1, \ldots, r_k\},$$

with

$$0 = r_0 < r_1 < \cdots < r_K = M, \qquad |G_k| = g_k = r_k - r_{k-1}.$$

The group size is chosen so that the group working set fits in cache:

$$g_k \, s \, N_p \le C_{\text{cache}}, \qquad g_k \le \left\lfloor \frac{C_{\text{cache}}}{s N_p} \right\rfloor.$$

Thus a row group is scanned while its data remains cache-resident.

SIMD column blocks

For cache line $\ell = 1, \ldots, L$ and SIMD block $q = 1, \ldots, V$, define the column indices

$$j_{\ell,q,p} = (\ell - 1)E + (q - 1)w + p, \qquad p = 1, \ldots, w.$$

Let the corresponding SIMD load be

$$\mathbf{a}_{i,\ell,q} = \begin{bmatrix} A_{i,j_{\ell,q,1}} \\ A_{i,j_{\ell,q,2}} \\ \vdots \\ A_{i,j_{\ell,q,w}} \end{bmatrix} \in \mathbb{R}^w.$$

The tail mask is

$$\mathbf{m}_{\ell,q} = \begin{bmatrix} m_{\ell,q,1} \\ \vdots \\ m_{\ell,q,w} \end{bmatrix}, \qquad m_{\ell,q,p} = \begin{cases} 1, & j_{\ell,q,p} \le N, \\ 0, & j_{\ell,q,p} > N. \end{cases}$$

Vectorized row and group averages

For row i, accumulate the masked SIMD vectors:

$$\mathbf{s}_i = \sum_{\ell=1}^{L} \sum_{q=1}^{V} \mathbf{a}_{i,\ell,q} \odot \mathbf{m}_{\ell,q}.$$

The row average is the horizontal reduction

$$\mu_i = \frac{1}{N} \sum_{p=1}^{w} [\mathbf{s}_i]_p.$$

For group G_k, accumulate the row accumulators:

$$\mathbf{s}_{G_k} = \sum_{i \in G_k} \mathbf{s}_i.$$

The average over all columns in the rows of G_k is

$$\overline{A}_k = \frac{1}{N g_k} \sum_{p=1}^{w} [\mathbf{s}_{G_k}]_p = \frac{1}{g_k} \sum_{i \in G_k} \mu_i.$$

If the column-wise average vector over the group is also required, define

$$\mathbf{c}_{k,\ell,q} = \sum_{i \in G_k} \mathbf{a}_{i,\ell,q}, \qquad \mu_{k,j_{\ell,q,p}} = \frac{[\mathbf{c}_{k,\ell,q}]_p}{g_k},$$

for $j_{\ell,q,p} \le N$. Equivalently,

$$\mu_{k,j} = \frac{1}{g_k} \sum_{i \in G_k} A_{i,j}, \qquad \overline{A}_k = \frac{1}{N} \sum_{j=1}^{N} \mu_{k,j}.$$

Blocked scan

for $k = 1, \ldots, K$:
  $\mathbf{s}_{G_k} \leftarrow \mathbf{0}$
  for $i \in G_k$:
    $\mathbf{s}_i \leftarrow \mathbf{0}$
    for $\ell = 1, \ldots, L$:
      for $q = 1, \ldots, V$:
        $\mathbf{a} \leftarrow \mathrm{SIMD_LOAD}(A_{i,j_{\ell,q,1}}, \ldots, A_{i,j_{\ell,q,w}})$
        $\mathbf{s}_i \leftarrow \mathbf{s}_i + \mathbf{a} \odot \mathbf{m}_{\ell,q}$
    $\mu_i \leftarrow \dfrac{1}{N} \displaystyle\sum_{p=1}^{w} [\mathbf{s}_i]_p$ (if per-row output is needed)
    $\mathbf{s}_{G_k} \leftarrow \mathbf{s}_{G_k} + \mathbf{s}_i$
  $\overline{A}_k \leftarrow \dfrac{1}{N g_k} \displaystyle\sum_{p=1}^{w} [\mathbf{s}_{G_k}]_p$

Cost model

Each row touches $L = \lceil N/E \rceil$ cache lines, so the total memory traffic is

$$T = \sum_{k=1}^{K} g_k L = M \left\lceil \frac{N}{E} \right\rceil \quad \text{cache lines.}$$

The number of SIMD vector operations is approximately

$$I = \sum_{k=1}^{K} g_k L V = \frac{MN}{w} \quad \text{vector loads/adds,}$$

plus $O(M)$ horizontal reductions if every row average is needed. Division is deferred until after accumulation, so the inner scan uses only vector loads, masks, and additions.